

Show all steps or explain how you found your answer. Use digital inking for each question involving mathematical symbols. If the answer is a sentence with no mathematical symbols (like question 3), feel free to type your response to that question. However, do not mix digital inking with typing.

1. Write the following statements as a combination of p and q using propositional logic where

p: I will do all of my homework

q: I will get an A in Discrete Math

- a. I will get an A in Discrete Math **only if** I do all of my homework (5 pts)
- b. **Either** I will not get an A in Discrete Math **or** I will **not** do my homework (5 pts)
- c. **Either if** I **don't** do all of my homework, **then** I **won't** get an A in Discrete Math, **or** I will do all of my homework **and** get an A in Discrete Math. (5 pts)

2. Show that $\neg(\neg p \vee q) \equiv (\neg q \wedge p)$

- a. using a truth table (5 pts)

- b. using logical equivalences (5 pts)

3. What is the converse, inverse and contrapositive of the statement “If I get a tax refund, then I will go on vacation”?
- a. Assign each proposition and write the original statement using those propositions. (5 pts)
 - b. Converse (5 pts)
 - c. Inverse (5 pts)
 - d. Contrapositive (5 pts)
4. Suppose that $Q(x)$ is the propositional function “ $5x + 2 = x$.” What are the truth values for $\forall x Q(x)$ and $\exists x Q(x)$ for the domain of all real numbers? Explain! (5 pts)
5. Prove or disprove each of the following statements. Use any method. (10 pts each)
- a. The sum of two odd integers is always even.

b. The product of an even integer and an even integer is always even.

c. 10 is not the square of a positive integer (Be sure your proof covers ALL integer values of x . To do this, use proof by cases where *Case 1 is $1 \leq x \leq 3$ and Case 2 is $x \geq 4$*)

d. There are no solutions in positive integers to the equation
 $x^4 + y^4 = 1000$.

6. Use rules of inference to show that the premises $p \vee q, q \rightarrow r, r \rightarrow s, \neg p$ lead to a conclusion of s .

Bonus (5 pts)

Give the output, Q , of the following logic gate, including all intermediate steps.

